THE CATEGORY OF DOTTED ARROWS IS INDEPENDENT OF THE CHOSEN 2-MORPHISM WHEN THE INERTIA OF THE TARGET IS PROPER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The situation of the valuative criterion with its category of dotted arrows and the choice of a 2-morphism filling the square. Referred to in the statement (source lines 9152-9158) as a pointer to the defining definition of this chapter. Fixes the categories whose equivalence is asserted.
- The valuative criterion of properness for morphisms of algebraic spaces. Invoked inside the counted proof (source lines 9175-9176) as a cross-chapter pointer. Produces an automorphism of the object over the valuation ring restricting to the given automorphism over the fraction field.

Source reference. Original definition definition-fill-in-diagram: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 9116–9116. Navigation pointer only: the counted text refers to the Stacks item definition-fill-in-diagram, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-characterize-proper: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 9176–9176. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-characterize-proper, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *In Definition definition-fill-in-diagram assume $\mathcal{I}_\mathcal{Y} \to \mathcal{Y}$ is proper (for example if $\mathcal{Y}$ is separated or if $\mathcal{Y}$ is separated over an algebraic space). Then the category of dotted arrows is independent (up to noncanonical equivalence) of the choice of γ and the existence of a dotted arrow (for some and hence equivalently all γ) is equivalent to the existence of a diagram*

$$\begin{array}{ccc} \mathrm{Spec}(K) & \xrightarrow{x} & \mathcal{X} \\ {\scriptstyle j}\downarrow & \nearrow_{a} & \downarrow{\scriptstyle f} \\ \mathrm{Spec}(A) & \xrightarrow{y} & \mathcal{Y} \end{array}$$

with 2-commutative triangles (without checking the 2-morphisms compose correctly).

Proof. Let $\gamma, \gamma' : y \circ j \longrightarrow f \circ x$ be two 2-morphisms. Then $\gamma^{-1} \circ \gamma'$ is an automorphism of y over $\mathrm{Spec}(K)$. Hence if $Isom_\mathcal{Y}(y, y) \to \mathrm{Spec}(A)$ is proper, then by the valuative criterion of properness (Morphisms of Spaces, Lemma spaces-morphisms-lemma-characterize-proper) we can find $\delta : y \to y$ whose restriction to $\mathrm{Spec}(K)$ is $\gamma^{-1} \circ \gamma'$. Then we can use δ to define an equivalence between the category of dotted arrows for γ to the category of dotted arrows for γ' by sending (a, α, β) to $(a, \alpha, \beta \circ \delta)$. The final statement is clear. □